

The Model Scheduling of Distributed Inference Network (DIN)

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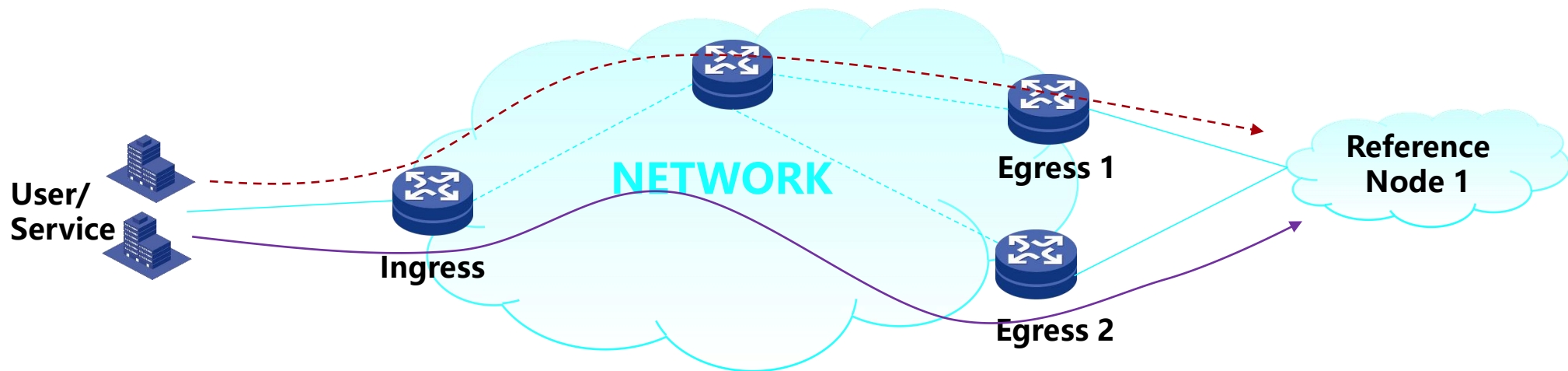
Distributed Model Scheduling

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Model Scheduling Technology Value

Why does DIN need the Model Scheduling?

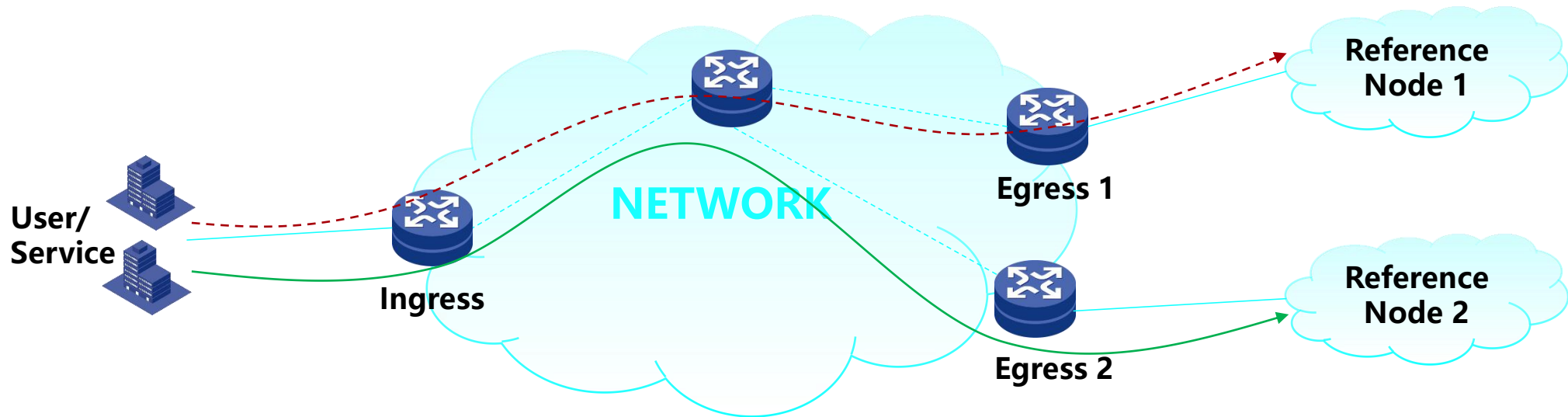
- Computing resources (such as GPU/TPU) and storage resources (such as model weights) in a distributed network are typically heterogeneous and **dynamically changing**. Model scheduling is needed to avoid resource overload (e.g., GPU out-of-memory errors) or resource idling (e.g., some nodes being idle while others are overloaded).
- Large models (such as LLMs with hundreds of billions of parameters) cannot be fully deployed on a single node and need to be divided into multiple sub-modules (e.g., by layer or by computation graph), which are then distributed across different nodes. Model scheduling is required to coordinate the execution order of these sub-modules.
- The time data spends in transmission over the network and waiting for computing resources can significantly impact real-time requirements (such as autonomous driving and online customer service). It is necessary to select the node with the lowest latency through model scheduling.
- Different nodes may be equipped with different hardware (e.g., NVIDIA GPU, Intel NPU, ASIC), and the execution environment for the model needs to be adapted through scheduling.
- For one node, Node failures or traffic bursts may lead to inference disruptions or a decline in quality of service, and High-concurrency inference may consume significant energy and computing resources, increasing operation expenditure.



What is the Model Scheduling in DIN?

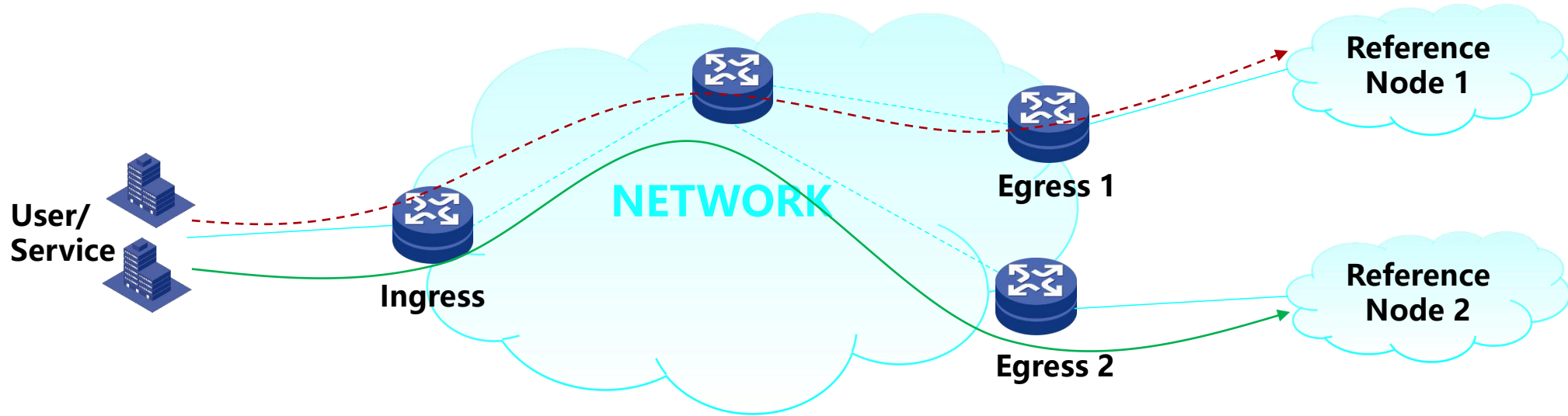
The Model Scheduling in DIN:

- Based on multi-dimensional factors such as user requirements, device performance, and network status;
- Aims to dynamically allocate and optimize model resources;
- To achieve efficient and low-latency model inference services.



Core goal : maximize resource utilization and reduce operational costs under the premise of guaranteed quality of service (QoS).

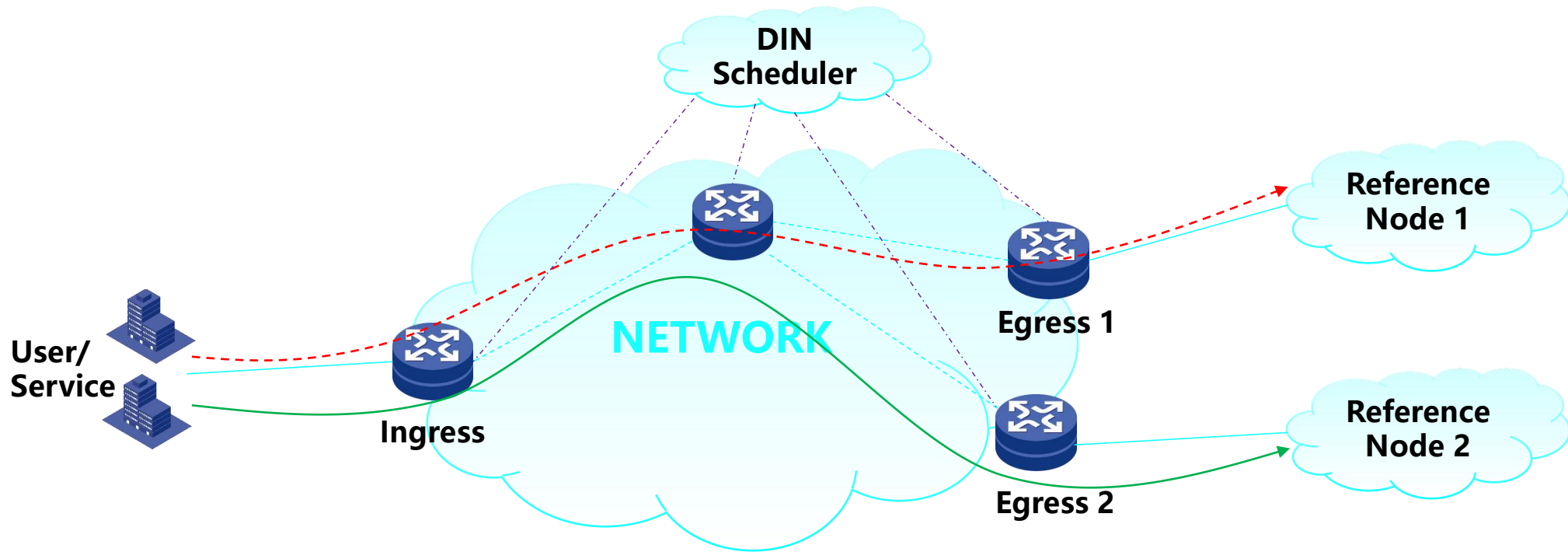
The Model Scheduling in DIN



How to do the Model Scheduling :

- ◆ When a service or user requests an inference service, the edge ingress node performs resource scheduling based on the service or user 's inference requirements to obtain the optimal reference node.
- ◆ Through efficient model scheduling, The inference request maybe be forwarded to Cloud 1 model or Cloud 2 model for AI inference.

The Model Scheduling Schemes

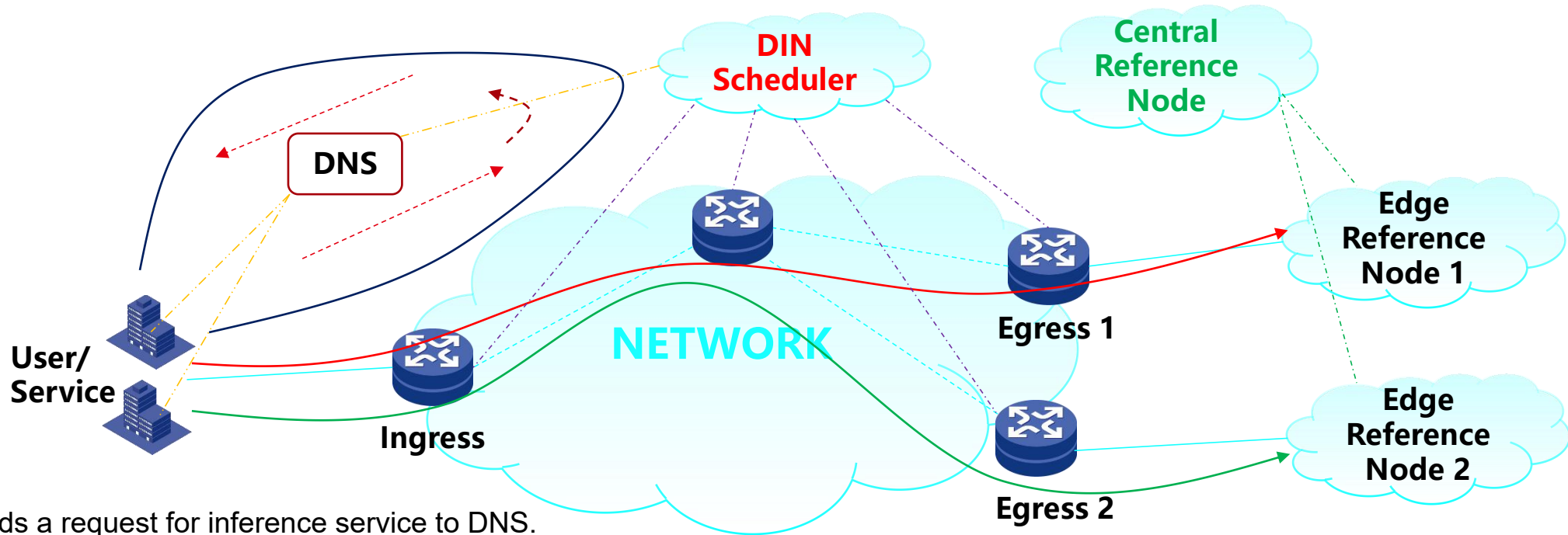


Two Model Scheduling techniques for inference services :

Centralized Model Scheduling : the DIN Scheduler is used to select model inference nodes.

Distributed Model Scheduling : the Computing-Aware Traffic Steering (CATS) is used to select model inference nodes.

The Centralized Model Scheduling

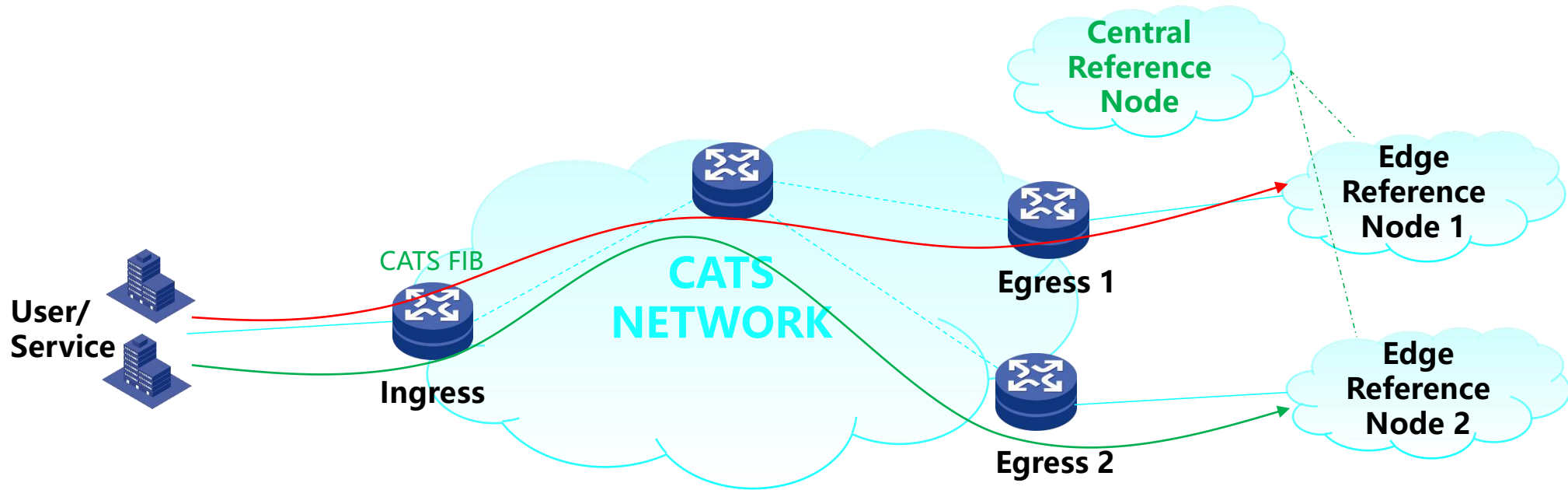


- ✓ User sends a request for inference service to DNS.
- ✓ DNS domain name resolution first resolves to the DIN Scheduler. Then the user request is sent to the DIN Scheduler.
- ✓ The **DIN Scheduler** assigns the optimal inference node IP address **based on principles** such as edge node load, proximity to user IP addresses, network quality, and traffic balancing.
- ✓ The optimal inference node IP address is returned to the DNS, and then send it back to the User.
- ✓ The user re-initiates the inference request to directly access the optimal edge inference node. And the inference result may be directly returned by the optimal edge inference node, or collaborating with the central inference node for improved reasoning accuracy.

What to do next ?

How the centralized scheduling algorithm within the DIN scheduler satisfies varying user requirements ?
How to reduce centralized scheduling response time, particularly for the initial packet ?

The Distributed Model Scheduling

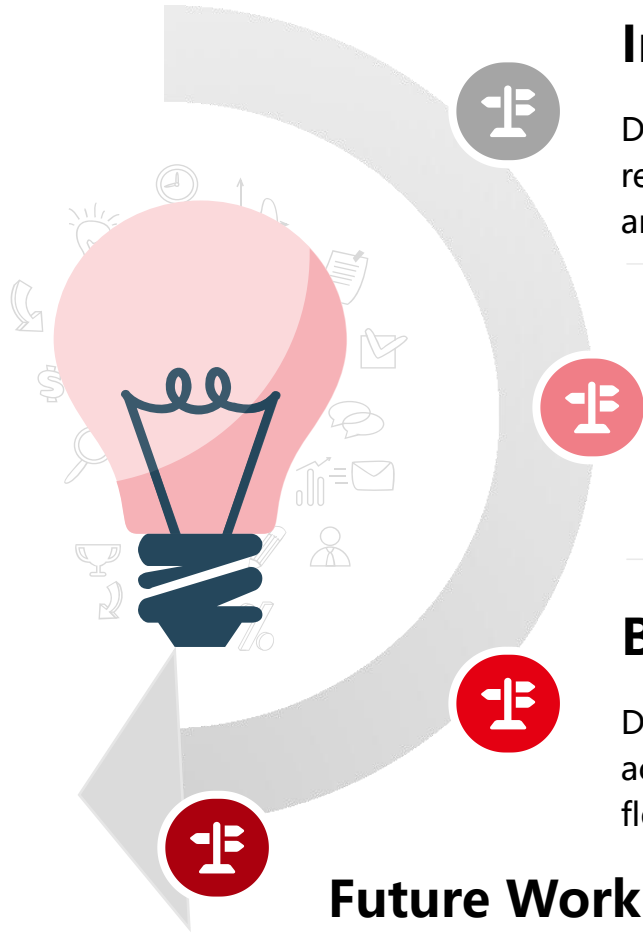


- ✓ Use the Computing-Aware Traffic Steering (CATS) network, which supports compute awareness, advertisement, and joint routing functions. For more detailed information, see the CATS WG drafts, such as draft-ietf-cats-framework.
- ✓ By perceiving deployment locations, real-time states, and load information of computing resources/services, as well as inference service demands.
- ✓ Dynamically generates on-demand scheduler policies to dispatch service request along optimal network paths to target inference nodes, Based on the multi-factor joint scheduling algorithm of "computing power + network".
- ✓ By searching the CATS Forwarding Information Table (FIB) at the DIN ingress node, the inference request is scheduled to the optimal reference node.

What to do next ?

How to appropriately extend Routing Protocol to transmit computing power information ? BGP (Origin Address Family or New Address Family).

Technology Value of Model Scheduling



Intelligent Resource Allocation

DIN scheduling technology realizes intelligent and dynamic resource allocation, significantly improving resource utilization and reducing enterprise operation cost.

High-Availability, Low-Latency Service Assurance

Through centralized and distributed scheduling schemes, DIN can provide users with highly available, low-latency AI inference services in complex environments, enhancing the user experience.

Business Flexibility and Differentiation Empowerment

DIN flexibly responds to diverse inference requirements (such as different inference accuracy levels), empowering carriers to achieve service differentiation and business flexibility expansion, thus strengthening market competitiveness.

Future Work

Improve the Model Scheduling method to satisfy varying user requirements and reduce the scheduling response time.
Combine the DIN model scheduling with the CATS network.

THANKS !